

DIEGO RABELO DA COSTA

CURRICULUM SUMMARY

CONTACT

Condensed Matter Theory Group (GTMC)
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Google Scholar: scholar.google.com/citations?user=GqQD7J4AAAAJ | GitHub: github.com/drabelocosta



EDUCATION

May 2018 - Jan. 2019 Postdoctoral researcher (Research Scholar) of UofM and HU

Institutions: Tony Low's Lab, Department of Electrical & Computer Engineering, University of Minnesota, USA, and Chujun Zhao's Lab, School of Physics & Electronics, Hunan University, Changsha, Hunan, China
Joint post-doc position: bilateral postdoctoral agreement between Prof. Chujun Zhao's lab at HU and Prof. Tony Low's lab at UofM.

Dec. 2014 - Jan. 2018 Postdoctoral researcher (PDJ/CNPq and PNPd/CAPES)

Institution: Department of Physics, Federal University of Ceará, Brazil. Visiting researcher at the University of Antwerp (Belgium, ≈ 5 months in 2015) and at the University of Minnesota (USA, 2016 and 2017).

2011 - 2014 Ph.D. in physics

Institutions: Department of Physics, Federal University of Ceará, Brazil and Department of Physics, University of Antwerp, Belgium
Title: Transport and confinement in monolayer and bilayer graphene nanostructures with different edges, interfaces, and potentials
Promoter: Gil de Aquino Farias | Co-promoter: Andrey Chaves and François M. Peeters
Joint Ph.D.: Exchange period (1 year) at the University of Antwerp financially supported by the Brazilian Program Science Without Borders and CAPES Foundation

2010 - 2011 Master in physics

Institution: Department of Physics, Federal University of Ceará, Brazil
Title: Ideal semiconductor quantum rings
Promoter: Gil de Aquino Farias | Co-promoter: Andrey Chaves

2006 - 2009 Bachelor in physics

Institution: Department of Physics, Federal University of Ceará, Brazil
Title: Effects of impurities and electric fields on the energy spectrum of ideal semiconductor quantum rings
Promoter: Jeanlex Soares de Sousa

PROFESSIONAL HISTORY

- **Fundação Cearense de Apoio ao Desenvolvimento Científico e Tecnológico (FUNCAP), Ceará, Brazil**

On secondment (“cessão”) from UFC to the Ceará State Research Foundation since February 2025.

03/Feb. 2025 – 01/Oct. 2025 *Manager (symbol DNS-3) of the Management of Analysis and Evaluation of Studies (GERAE).*

01/Oct. 2025 – 20/May 2026 *Special Advisor to the FUNCAP’s Presidency (symbol DNS-2).*

20/May 2026 – Present *Director (symbol DNS-2) of the Internal Control, Ombudsman and Social Communication Advisory Office (ASCOI).*

19/June 2026 – Present *FUNCAP’s Ombudsman (“Ouvidor”), by designation (Portaria nº 0031/2026).*

- **Federal University of Ceará (UFC), Fortaleza, Brazil**

Sep. 2014 – Dec. 2014 *Teaching Assistant*

Teaching assistant for undergraduate courses in Chemistry, including theoretical classes.

Feb. 2019 – Present *Adjunct Professor (Professor do Magistério Superior)*

Department of Physics, Condensed Matter Theory Group (GTMC); undergraduate (Fundamental and Experimental Physics for Engineering) and graduate (Advanced Topics, Electronic Transport in Mesoscopic Systems) courses; permanent member of the Graduate Program in Physics.

- **Conselho Nacional de Desenvolvimento Científico e Tecnológico (CNPq), Brazil**

Feb. 2019 – Present *Research Productivity Fellow (Bolsista PQ, level 2)*

- **Referee** for Physical Review B, Journal of Physics: Condensed Matter, New Journal of Physics, European Physical Journal Plus, Superlattices and Microstructures, Journal of Computational Electronics, and Solid State Communications.

LIST OF UP TO 5 OF THE MOST CITED SCIENTIFIC PUBLICATIONS

Citation counts from Google Scholar, October 2025.

- [1] A. Chaves, J. G. Azadani, H. Alsalman, D. R. da Costa, R. Frisenda, A. J. Chaves, S. H. Song, Y. D. Kim, D. He, J. Zhou, A. Castellanos-Gomez, F. M. Peeters, Z. Liu, C. L. Hinkle, S.-H. Oh, P. D. Ye, S. J. Koester, Y. H. Lee, Ph. Avouris, X. Wang, and T. Low, *Bandgap engineering of two-dimensional semiconductor materials*, npj 2D Materials and Applications **4**(1), 29 (2020). **1181 citations**
- [2] D. R. da Costa, A. Chaves, M. Zarenia, J. M. Pereira Jr, G. A. Farias, and F. M. Peeters, *Geometry and edge effects on the energy levels of graphene quantum rings: a comparison between tight-binding and simplified Dirac models*, Physical Review B **89**, 075418 (2014). **89 citations**
- [3] D. J. P. de Sousa, L. V. de Castro, D. R. da Costa, J. M. Pereira Jr, and T. Low, *Multilayered black phosphorus: From a tight-binding to a continuum description*, Physical Review B **96**, 155427 (2017). **85 citations**
- [4] D. R. da Costa, A. Chaves, S. H. R. Sena, G. A. Farias, and F. M. Peeters, *Valley filtering using electrostatic potentials in bilayer graphene*, Physical Review B **92**, 045417 (2015). **71 citations**
- [5] D. R. da Costa, M. Zarenia, A. Chaves, G. A. Farias, and F. M. Peeters, *Analytical study of the energy levels in bilayer graphene quantum dots*, Carbon **78**, 392 (2014). **43 citations**

ACADEMIC QUANTITATIVE INDICATORS

- 90 publications in peer-reviewed journals (42 Phys. Rev. B; 10 J. Phys.: Condens. Matter; 8 Physica E; 4 Nanotechnology; 3 J. Appl. Phys.; 3 Phys. Rev. E; 2 Nano Lett.; 1 Nat. Commun.; 1 Phys. Rev. Lett.; 1 npj 2D Mater. Appl.; 1 2D Mater.; 1 Nanophotonics; 1 Carbon; and others), plus 3 preprints under review
- ≈ 3130 citations, h-index 24 and i10-index 49 (Google Scholar, October 2026)
- Supervising dissertations (Lattes CV, April 2026)

- Guidance and supervision in progress:
 - * Postdoctoral researchers: 1 as Supervisor
 - * Ph.D. students: 8 as Promoter and 1 as Co-Promoter
 - * Master students: 2 as Co-Promoter
 - * Undergraduate research students: 1 as Promoter
- Completed orientations and supervision:
 - * Postdoctoral researchers: 1 as Supervisor
 - * Ph.D. students: 9 as Co-Promoter
 - * Master students: 10 as Promoter and 8 as Co-Promoter
 - * Bachelor/undergraduate research students: 7 as Promoter
- Scientific output: Conferences/Workshops, Examination boards, and Scientific Visits
 - Member of 131 examination boards: 29 Ph.D. theses; 36 master dissertations; 43 qualifying exams (master and Ph.D.); 20 bachelor monographs; 3 other juries
 - Member of 6 organizing committees of scientific events (e.g., 19th, 20th and 21st Brazilian Workshop on Semiconductor Physics; SBF Autumn Meetings 2019 and 2024)
 - 41 participations in national and international scientific events, including 8 invited talks

RESEARCH INTERESTS

- Two-dimensional materials; - Van der Waals materials and heterostructures; - Flat-band lattices (Lieb, kagome and α - T_3); - Moiré and twisted bilayers (twistronics); - Excitons and trions in 2D semiconductors; - Semiconductor heterostructures; - Materials modelling (tight-binding, continuum/ $\mathbf{k}\cdot\mathbf{p}$ models, split-operator technique, transfer matrix); - Electronic, transport, magnetic and optical properties of nanostructured systems; - Plasmonics; - Statistical physics of magnetic systems.

HONOURS AND AWARDS

- CNPq Research Productivity Fellowship (PQ-2) since 2019
- The paper “Position-dependent mass effects on the electronic states of a quantum torus” was highlighted as an *Editor’s Choice* selection of Physica E, volume 184 (2026).
- 2 Best Poster Awards. Section of Semiconductors. XXXVI (2013) and XXXV (2012) Brazilian Condensed Matter Physics Meetings. Awarded by the Brazilian Physical Society
- A picture of one of my papers was selected for the Physical Review B Kaleidoscope [Phys. Rev. B **92**(12), 125441 (2015)]
- The paper “Magnetic field induced vortices in graphene quantum dots” was featured on the cover of the Journal of Physics: Condensed Matter in the issue of April 10th, 2020.